

6 (a) (i)	electron	B1	[1]
(ii)	any two : can be deflected by electric and magnetic fields or negatively charged / absorbed by few (1 – 4)mm of aluminum / 0.5 to 2 m or metres for range in air / speed up to 0.99c / range of speeds / energies	B2	[2]
(iii)	decay occurs and cannot be affected by external / environmental factors or two stated factors such as chemical / pressure / temperature / humidity	B1	[1]
(b)	3 and 0 for superscript numbers 2 and –1 for subscript numbers	B1 B1	[2]
(c)	energy = $5.7 \times 10^3 \times 1.6 \times 10^{-19}$ (= 9.12×10^{-16} J)	C1	
	$v^2 = \frac{2 \times 9.12 \times 10^{-16}}{9.11 \times 10^{-31}}$	C1	
	$v = 4.5 \times 10^7 \text{ ms}^{-1}$	A1	[3]
(d)	both have 1 proton and 1 electron 1 neutron in hydrogen-2 and 2 neutrons in hydrogen-3 (special case: for one mark 'same number of protons / atomic number different number of neutrons')	B1 B1	[2]